

Certifying Γ -Robust Discrete Pricing via Full-Breakpoint Multiple-Choice Knapsack Decomposition

V3 Legacy Reconstruction

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Archival status. The original V3 manuscript source and compiled PDF were stored only in an ignored local artifact directory and are no longer available. This document is a transparent legacy reconstruction from the exact arXiv v1 source of the preceding paper, the final V3 code snapshot at Git commit 2e2829e55243b846e5944965a839eb9b28dd5e9c, and that snapshot's README, REVISION_HISTORY, SUBMISSION, tests, and experiment drivers. It is not represented as a byte-identical copy of the lost manuscript. Statements unsupported by preserved outputs are identified as scope limits rather than recreated numerical claims.

Abstract

We reconstruct the final scientific position of the V3 robust discrete-pricing research line. A finite price menu with a portfolio margin requirement reduces exactly to a multiple-choice knapsack problem. Under an integer cardinality-budget uncertainty set, the coupled robust margin constraint is equivalent to a finite union of fixed-threshold MCKPs over the complete set of original deviation breakpoints. For every threshold, upper-hull geometry gives an exact LP bound and a one-item additive rounding certificate. A globally certifying branch-and-bound must preserve original integer options, initialize a valid upper-bound record for every threshold, and validate every incumbent against the original robust constraint. The V3 implementation further provides an exact increasing-threshold sweep that updates transformed coefficients while reusing a hull only when its point set is unchanged. This reconstruction states the corresponding validity invariants, separates exact procedures from diagnostic reductions, and records the limits of the surviving empirical evidence. Its purpose is archival continuity between the public March 2026 paper and the independent V4 group-envelope pivot.

Keywords: robust optimization; discrete pricing; multiple-choice knapsack; exact decomposition; parametric sweep

1 Research position and provenance

The V3 line studies a discrete portfolio-pricing model in which one price is selected from each product menu. Portfolio revenue is maximized subject to a minimum margin ratio and price-admissibility restrictions. The March 2026 paper established the exact reduction to a multiple-choice knapsack problem

(MCKP), the fixed-threshold LP geometry, and an additive one-item rounding bound. V3 shifted the center of gravity from a fast heuristic narrative to a certifying decomposition framework:

1. enumerate the complete original deviation-breakpoint set;
2. construct a valid fixed-threshold LP upper bound at every breakpoint;
3. maintain globally valid lower and upper bounds under time or node limits;
4. preserve below-hull integer options in exact search;
5. update the fixed-threshold data in an exact parametric sweep; and
6. treat reduced threshold sets and local-budget extensions as separately scoped diagnostics unless their own exactness conditions hold.

This is not a claim that threshold decomposition, MCKP hull geometry, or Lagrangian relaxation is new. The contribution of the V3 program is the combination of the pricing reduction, explicit robust and rounding certificates, and defensible global-gap accounting in a reproducible solver.

2 Discrete pricing and MCKP reduction

Let $i \in \{1, \dots, n\}$ index products and J_i the admissible finite menu of product i . Option (i, j) has price $p_{ij} > 0$, exposure $w_i \geq 0$, reference price $\bar{p}_i > 0$, and nominal demand $\hat{g}_{ij} \geq 0$. Binary variable x_{ij} selects one option:

$$\sum_{j \in J_i} x_{ij} = 1, \quad x_{ij} \in \{0, 1\}.$$

Define nominal revenue and reference-denominator coefficients

$$v_{ij} := w_i p_{ij} \hat{g}_{ij}, \quad q_{ij} := w_i \bar{p}_i \hat{g}_{ij}.$$

For a required margin ratio $\Delta > 0$, nominal feasibility is

$$\sum_{i,j} (v_{ij} - \Delta q_{ij}) x_{ij} \geq 0.$$

Writing $s_{ij} := v_{ij} - \Delta q_{ij}$ and $s_i^* := \max_j s_{ij}$ gives nonnegative transformed costs $c_{ij} := s_i^* - s_{ij}$ and capacity $C := \sum_i s_i^*$. Thus the nominal model is exactly

$$\begin{aligned} \max_x \quad & \sum_{i,j} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i,j} c_{ij} x_{ij} \leq C, \\ & \sum_j x_{ij} = 1 \quad (i = 1, \dots, n), \quad x \in \{0, 1\}. \end{aligned}$$

Proposition 1 (Exact nominal reduction). *The baseline transformation preserves every feasible price selection and its objective value. Hence the finite-menu pricing model with a positive denominator and a single portfolio ratio floor is an MCKP.*

Proof. For every exactly-one selection, $\sum_{i,j} s_{ij}x_{ij} = \sum_i s_i^* - \sum_{i,j} c_{ij}x_{ij}$. The margin residual is nonnegative if and only if the transformed cost is at most C . The objective and group equations are unchanged. $\square$

3 Budgeted uncertainty and full breakpoints

Let $t_{ij} \in \mathbb{R}$ be the signed option-level margin exposure produced by an adverse demand perturbation, and set $d_{ij} := |t_{ij}|$. Under an integer budget $\Gamma \in \{0, \dots, n\}$, at most Γ selected groups receive their full adverse deviation. The robust margin residual is

$$\sum_{i,j} s_{ij}x_{ij} - \max_{\substack{0 \leq u_i \leq 1 \\ \sum_i u_i \leq \Gamma}} \sum_i u_i \sum_j d_{ij}x_{ij}.$$

LP duality for the inner budget problem yields

$$\max_{0 \leq u \leq 1, \mathbf{1}^\top u \leq \Gamma} \sum_i d_i u_i = \min_{\theta \geq 0} \left\{ \Gamma\theta + \sum_i [d_i - \theta]_+ \right\}, \quad d_i := \sum_j d_{ij}x_{ij}.$$

For fixed θ , define

$$s_{ij}^\theta := s_{ij} - [d_{ij} - \theta]_+.$$

The complete original candidate set is

$$\mathcal{B} := \{0\} \cup \{d_{ij} : i = 1, \dots, n, j \in J_i\}.$$

Theorem 2 (Finite full-breakpoint decomposition). *A binary selection is robust feasible if and only if there exists $\theta \in \mathcal{B}$ such that*

$$\sum_{i,j} s_{ij}^\theta x_{ij} - \Gamma\theta \geq 0.$$

Consequently, the robust problem is the maximum over a finite family of fixed- θ MCKPs.

Proof. For a fixed selection, the protection function $h(\theta) = \Gamma\theta + \sum_i [d_i - \theta]_+$ is convex and piecewise linear. Its breakpoints are zero and the selected deviations. A minimizer therefore exists at one of those values, all of which belong to the global set $\mathcal{B}$. Substitution into the robust residual gives the stated equivalence. $\square$

The word *original* is essential. Removing an option before constructing $\mathcal{B}$ can remove the only threshold supporting an optimal integer selection. V3 therefore permits reduced threshold sets only as diagnostics or heuristics, not as inputs to a global exactness claim.

4 Fixed-threshold hull certificate

For fixed $\theta \in \mathcal{B}$, let

$$s_i^*(\theta) := \max_j s_{ij}^\theta, \quad c_{ij}^\theta := s_i^*(\theta) - s_{ij}^\theta, \quad C(\theta) := \sum_i s_i^*(\theta) - \Gamma\theta.$$

If $C(\theta) < 0$, the disjunct is infeasible. Otherwise it is the MCKP

$$\begin{aligned} L(\theta) := \max_x \quad & \sum_{i,j} v_{ij} x_{ij} \\ \text{s.t.} \quad & \sum_{i,j} c_{ij}^\theta x_{ij} \leq C(\theta), \\ & \sum_j x_{ij} = 1, \quad x \geq 0. \end{aligned}$$

Within a group, plot each option as (c_{ij}^θ, v_{ij}) . Dominated points do not affect the LP relaxation. The upper concave hull supplies marginal value-per-cost segments; sorting all segments by decreasing slope gives the exact greedy solution of the LP relaxation. At most one group is fractional after merging equal-slope ties in a consistent way.

Let ΔV_i^θ be the largest value increase between consecutive vertices of group i 's upper hull, and let $\Delta V_{\max}^\theta := \max_i \Delta V_i^\theta$.

Theorem 3 (One-item additive certificate). *For a feasible fixed threshold, let $Z^{\text{IP}}(\theta)$ and $L(\theta)$ be the integer and LP optima. Then*

$$0 \leq L(\theta) - Z^{\text{IP}}(\theta) \leq \Delta V_{\max}^\theta.$$

A HullRound solution obtained by rounding the sole fractional group toward the lower-cost adjacent hull vertex is feasible and attains value at least $L(\theta) - \Delta V_{\max}^\theta$.

Proof. The greedy LP fills complete hull segments until the residual capacity cuts at most one segment. Rounding that segment down cannot increase cost. The value loss is no larger than the complete value jump of that segment, which is at most $\Delta V_{\max}^\theta$. The integer optimum lies between the rounded feasible value and the LP upper bound. $\square$

Corollary 4 (Asymptotic relative gap). *If option values and hull jumps are uniformly bounded while the optimal value grows linearly in n , then the relative fixed-threshold integrality gap is $O(1/n)$.*

Upper-hull filtering is safe for the LP calculation but not automatically for the integer search. A below-hull option can remain part of an optimal integer combination because capacities are discrete. V3's exact branch-and-bound keeps all original options except those removed by an integer-safe within-group dominance rule.

5 Globally valid exact and anytime search

For every $\theta \in \mathcal{B}$, denote the fixed-threshold integer optimum by $Z^{\text{IP}}(\theta)$. The robust optimum is

$$Z^{\text{robust}} = \max_{\theta \in \mathcal{B}} Z^{\text{IP}}(\theta).$$

The exact solver maintains a robust-feasible incumbent value Z^{inc} and, for each threshold, a valid record U_θ satisfying $Z^{\text{IP}}(\theta) \leq U_\theta$. Initially, $U_\theta = L(\theta)$ whenever the fixed-threshold LP is feasible and $U_\theta = -\infty$ otherwise. Branch-and-bound may subsequently reduce U_θ or prove the disjunct optimal.

Proposition 5 (Global gap invariant). *If every original threshold has a valid upper-bound record, then at every stage of the search*

$$Z^{\text{inc}} \leq Z^{\text{robust}} \leq U^{\text{global}} := \max_{\theta \in \mathcal{B}} U_{\theta}.$$

Thus a time- or node-limited run reports the valid absolute gap $U^{\text{global}} - Z^{\text{inc}}$ and the corresponding scaled relative gap.

Proof. The incumbent is accepted only after direct validation under the original sorted- Γ robust residual, so it is a lower bound. Every robust feasible selection belongs to at least one original threshold disjunct. The optimum of that disjunct is bounded by its record, hence the maximum record bounds the global optimum. Branch-and-bound pruning preserves each record because it uses only valid LP bounds or proven infeasibility. $\square$

A finite limited-run gap is invalid if some threshold has no upper-bound record. This accounting rule was a non-negotiable V3 correction: initializing only the thresholds reached before a time limit can make an apparently small global gap meaningless.

6 Exact parametric threshold sweep

Independent recomputation constructs every array from scratch at every threshold. V3 also supplies an increasing- θ sweep. Between consecutive global breakpoints, each transformed coefficient $s_{ij}^{\theta} = s_{ij} - [d_{ij} - \theta]_{+}$ is affine; at a breakpoint, the corresponding option changes from the active-deviation expression to its nominal margin. The sweep updates these coefficients and then recomputes each group baseline, transformed cost array, and capacity exactly.

Hull reuse is deliberately conservative. A cached group hull is reused only when the nondominated option set and the complete cost-value point arrays are unchanged within the stated tolerance. Otherwise it is rebuilt. Optional validation independently reconstructs sampled states and compares transformed coefficients, capacities, costs, hulls, and root LP bounds.

Proposition 6 (Sweep equivalence). *In exact arithmetic, the increasing-threshold sweep produces the same s_{ij}^{θ} , c_{ij}^{θ} , $C(\theta)$, group hulls, and LP upper bound as independent fixed-threshold recomputation at every $\theta \in \mathcal{B}$.*

Proof. Each option’s update rule is the piecewise-linear identity defining s_{ij}^{θ} . Baselines, costs, and capacity are then evaluated from their definitions. A reused hull has an unchanged input point set; a changed point set is rebuilt. The same deterministic greedy merge therefore receives the same hull segments as independent recomputation. $\square$

This proposition is an exactness statement, not a universal runtime theorem. When most group point sets change at most thresholds, hull rebuilding and fixed-threshold integer search can dominate the saved coefficient work.

7 Implementation and preserved evidence

The complete code state is archived at commit 2e2829e55243b846e5944965a839eb9b28dd5e9c. The principal preserved modules are summarized in [Table 1](#).

Table 1. V3 code modules preserved in the exact archive.

Module	Role
<code>certificate.py</code>	Direct sorted- Γ robust-feasibility certificate.
<code>hull.py, greedy.py</code>	Group upper hulls and exact fixed-threshold LP relaxation.
<code>rounding.py, solver.py</code>	HullRound construction and one-item additive certificate diagnostics.
<code>exact_bnb.py</code>	Fixed-threshold and global full-breakpoint exact branch-and-bound with bound records.
<code>parametric_sweep.py</code>	Exact increasing-threshold state construction, guarded hull reuse, and validation diagnostics.
<code>local_budget.py</code>	Separately scoped segment-local budget extension for small threshold products.
<code>milp_baselines.py</code>	Optional SCIP, HiGHS, Gurobi, and CPLEX baselines with explicit availability reporting.

The V3 revision record states that 82 tests passed on 22 May 2026 and that public and blind manuscript builds passed at that time. It also records that the complete publication benchmark was not regenerated during the final revision. The generated CSVs, figures, and manuscript files lived in ignored local directories and did not survive. Accordingly, this reconstruction does not repeat unrecoverable V3 runtime tables or claim superiority over mature MILP solvers. The exact V2/arXiv numerical record remains preserved separately in the legacy archive.

The surviving design establishes a reproducible route to new evidence: unit tests exercise certificates and exact-search invariants; smoke drivers compare independent and swept states; solver scripts report unavailable backends rather than silently dropping them; and every accepted incumbent is checked against the original robust constraint.

8 Scope boundaries and relationship to V4

V3 is limited to integer Γ , one global budgeted margin constraint, and finite exactly-one menus. Fractional uncertainty budgets require an additional fractional protection term. Multiple independent robust constraints create a multidimensional threshold product and are not covered by the scalar decomposition. The pricing application uses own-price demand and does not establish causal or transaction-level demand effects.

V4 is not a revision of this paper. V3 studies fixed-threshold MCKP hull geometry, rounding, global enumeration, and exact-search accounting. V4 asks a different question: how to evaluate a Lagrangian multiplier simultaneously over the complete threshold family using exactly-one group envelopes. The two lines share classical robust-MCKP background but have distinct central theorems, algorithms, and evidence packages.

9 Conclusion

The V3 scientific contribution is best understood as a certifying bridge from finite-menu robust pricing to exact full-breakpoint MCKP decomposition. Its core guardrails are durable: use every original breakpoint for exactness, distinguish LP-safe hull filtering from integer-safe pruning, validate every incumbent in the original model, and report a finite global gap only after every disjunct has a valid upper-bound

record. The parametric sweep preserves these invariants while avoiding redundant state construction when group point sets remain unchanged.

This reconstructed PDF restores a readable V3 record without pretending to be the deleted original. The exact public predecessor, exact V3 code, provenance notes, checksums, and this reconstruction together form the maintained legacy archive.

Archived data and code

The exact code snapshot, predecessor source, this reconstruction source, and SHA-256 manifest are stored in the public legacy directory of https://github.com/eric939/robust_mckp. The V3 code is independently anchored by branch v3 and tag legacy-v3-code-20260522.

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